

Golden, CO (willing to relocate)
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Pratiksha Satish

[LinkedIn](#)
[Design Portfolio](#)
[GitHub](#)

EDUCATION

COLORADO SCHOOL OF MINES

Bachelor of Science in Design Engineering | Minor in Business & Entrepreneurship

Expected: May 2027

SKILLS & STRENGTHS

Technical: SolidWorks, Fusion 360, Autodesk Inventor (Certified), Python, Java, EES, Vensim, Rhino, ArcGIS

Modeling & Prototyping: Systems modeling, Dynamic simulation, Thermodynamic performance analysis

Communication & Leadership: Dialogue facilitation, Workshop facilitation, Event planning, Product marketing, Cross-functional collaboration, Project management

Languages: English (Fluent), French (Fluent), Tamil (Fluent)

EXPERIENCE

PEER EDUCATION

Peer Educator - Harm Reduction

Golden, CO

Aug 2025 - Present

- Led peer workshops and outreach events reaching 60+ students per event to promote mental health awareness
- Provided one-on-one support and connected students with appropriate campus wellness resources

UG RESEARCH

AR Design Team

Golden, CO

August 2026 - Present

- Review literature on robot redesign and AR/VR feedback elicitation techniques to establish HRI research foundations for a novel AI-based robot design tool
- Propose experimental design (research questions, hypotheses, methodology) and prototype robot morphologies/ environments in Unity to evaluate the tool with non-expert users

PROJECTS

Indoor Air Quality Systems Model

Aug 2025 - Dec 2025

- Modeled indoor PM2.5 accumulation and identified filtration and ventilation as primary drivers of long-term exposure levels
- Showed MERV 16 filtration cut long-term performance decline by about 65% at no added annual cost
- Simulated emissions policy reducing outdoor PM2.5 from 9 to $7 \mu\text{p}/\text{m}^3$, reducing indoor particle load about 40%

Combined Cycle Power Plant Performance Model

Oct 2025 - Dec 2025

- Built an EES model of a combined-cycle plant to evaluate power and efficiency impacts of system upgrades
- Modeled turbine inlet temperature increase, improving output about 470 kW with about 2.0 ROI
- Increased plant efficiency through enhanced HRSG heat recovery without additional fuel input

PVC Flute

March 2025 - May 2025

- Designed and tuned a playable PVC flute in the key of A using recycled materials and iterative acoustic testing

LEADERSHIP & INVOLVEMENT

SOUTH ASIAN STUDENT ASSOCIATION (SASA)

Golden, CO

Founder

Sep 2024 - Present

- Founded and serve as President of Mines' first South Asian Student Association, building it into a recognized campus presence
- Organized multiple cultural events attracting 100+ attendees per event while managing organization budget and cross-team partnerships with student organization

SOCIETY OF ASIAN SCIENTISTS & ENGINEERS (SASE)

Golden, CO

Festival Chair

May 2025 - Present

- Organized an annual Asian cultural festival with 300+ attendees, featuring performances, food, and cultural showcases
- Managed budgeting, subcommittees, and event logistics to ensure smooth execution

RELEVANT COURSEWORK:

Thermodynamics & Energy Systems (cycle analysis) · Systems Modeling & Simulation (stock-flow models, feedback loops) · Human-Centered Design (anthropometrics, inclusive design) · Physical Prototyping & Fabrication (arduino systems, iterative testing), GIS & Spatial Analysis